

Neuromuscular blocking agents, nondepolarizing: pancuronium

Hypokalaemia induced by glucocorticoids may enhance the blockade of nondepolarizing neuromuscular blocking agents, possibly leading to increased or prolonged respiratory depression or paralysis (apnea). Hydrocortisone has also been reported to decrease the efficacy of pancuronium by an unknown mechanism.

Potassium supplements

Effects of these medications and / or corticosteroids on serum potassium concentration may be decreased when used concurrently.

Ritodrine

Concurrent use may cause pulmonary oedema in the mother; maternal death has been reported; both medications should be discontinued at the first sign of pulmonary oedema.

Salicylates

Although concurrent use with glucocorticoids in the treatment of arthritis may provide additive therapeutic benefit and permit glucocorticoid dosage reduction, glucocorticoid may increase salicylate excretion and reduce salicylate plasma concentrations so that the salicylate dosage requirement may be increased; salicylism may occur when glucocorticoid dosage is subsequently decreased or discontinued, especially in patients receiving large (antirheumatic) doses of salicylates; also, the risk of gastrointestinal ulceration or hemorrhage may be increased.

Sodium - containing medication or foods

Concurrent use with pharmacologic doses of glucocorticoids may result in oedema and increased blood pressure, possibly to hypertensive levels.

Somatrem or Somatropin

Inhibition of the growth response to Somatropin or Somatrem may occur with chronic therapeutic use of daily doses (per square meter of body surface) in excess of parenteral hydrocortisone 5 to 7.5 mg. It is recommended that these doses not be exceeded during Somatropin or Somatrem therapy.

Streptozocin

Concurrent use with glucocorticoids may increase the risk of hyperglycaemia.

Vaccines, live virus or other immunizations

Administration of live virus vaccines to patients receiving pharmacologic (immunosuppressant) doses of glucocorticoids may increase the risk of the patient's developing the viral disease and / or decrease the patient's antibody response to the vaccine and is not recommended. Other immunizations are not recommended because of the increased risk of neurological complications and the possibility of decreased or absent antibody response.

Adequate protein intake

During prolonged therapy, a liberal protein intake is essential for counteracting the tendency to gradual weight loss sometimes associated with negative nitrogen balance, wasting and weakness of skeletal muscle.

Increased calcium excretion

All corticosteroids increase calcium excretion. Avoid abrupt cessation of corticosteroids.

Co-treatment with CYP3A inhibitors, including cobicistat-containing products, is expected to increase the risk of systemic side-effects. The combination should be avoided unless the benefit outweighs the increased risk of systemic corticosteroid side-effects, in which case patients should be monitored for systemic corticosteroid side-effects.

RECOMMENDED DOSAGE, DOSAGE SCHEDULE AND ROUTE OF ADMINISTRATION:

Dosage requirements are variable and must be individualized on the basis of the disease and the response of the patient.

Dexamethasone injection can be given without mixing or dilution, but if preferred, can be added without loss of potency to sodium chloride injection or dextrose injection or compatible blood for transfusion, and given by IV drip. The infusion mixture must be used within 24 hours and the usual aseptic techniques for injections should be observed.

All dosage recommendations are given in units of dexamethasone phosphate.

Usual adult and adolescent dose:

Intra-articular, intralesional, or soft-tissue injection:

In general, these injections are employed when only 1 or 2 joints or areas are affected.

Some of the usual single doses are found in the following table.

Site of Injection	Amount of Dexamethasone Phosphate
Large joints (e.g., knee)	2 - 4 mg (0.5 - 1 mL)
Small joints (e.g., interphalangeal, temporomandibular)	0.8 - 1 mg (0.2 - 0.25 mL)
Bursae	2 - 3 mg (0.5 - 0.75 mL)
Tendon sheaths *	0.4 - 1 mg (0.1 - 0.25 mL)
Soft-tissue infiltration	2 - 6 mg (0.5 - 1.5 mL)
Ganglia	1 - 2 mg (0.25 - 0.5 mL)
* Injection should be made into the tendon sheath, and not directly into the tendon.	

Frequency of injection: Repeated at 3-day to 3-week intervals, if necessary.

Intramuscular (IM) or intravenous (IV): 0.5 to 9 mg (0.125 to 2.25 mL) a day.

For cerebral oedema -

Initial: IV, 10 mg (2.5 mL), followed by 4 mg (1 mL) IM every 6 hours until symptoms subside. Dosage may be reduced after 2 to 4 days and gradually discontinued over a period of 5 to 7 days, unless a brain tumour, which must be treated before dexamethasone can be discontinued, is present.

Maintenance (for recurrent or inoperable brain tumours: IM, 2 mg (0.5 mL) 2 or 3 times a day initially, then adjusted according to patient response.

For shock -

The following regimens have been utilized:

IV, 20 mg (5 mL) as a single dose initially, followed by 3 mg (0.75 mL) per kg of body weight per 24 hours via continuous IV infusion, or

IV, 2 to 6 mg (0.5 to 1.5 mL) / kg body weight as a single injection, or

IV, 40 mg (10 mL) as a single dose, administered every 2 to 6 hours as needed, or

IV, 1 mg (0.25 mL) / kg body weight as a single injection.

Note: Administration of high-dose therapy for shock should be discontinued after the patient's condition has stabilized and is usually continued for no longer than 2 to 3 days.

Usual adult prescribing limits

Up to 80 mg (20 mL) daily.

Usual paediatric dose

Adrenocortical insufficiency - IM, 0.0233 mg (0.0058 mL) / kg body weight or 0.67 mg (0.1675 mL) / m² body surface a day (in 3 divided doses) every third day; or 0.00776 to 0.01165 mg (0.00194 to 0.00291 mL) / kg body weight or 0.233 to 0.335 mg (0.05825 to 0.08375 mL) / m² body surface once a day.

Other indications -

IM, 0.02776 to 0.16665 mg (0.00694 to 0.04166 mL) / kg body weight or 0.833 to 5 mg (0.20825 to 1.25 mL) / m² body surface every 12 to 24 hours.

Incompatibilities

Dexamethasone (as sodium phosphate) is physically incompatible with danorubicin, doxorubicin and vancomycin and should not be admixed with solutions containing these drugs. It also incompatible with doxapram hydrochloride and glycopyrrolate in a syringe.

SYMPTOMS AND TREATMENT FOR OVERDOSE AND ANTIDOTE(S):

Symptoms: Reports of acute toxicity and/or deaths following overdosage with glucocorticoids are rare. The adverse effects of corticosteroids are nearly always due to their use in excess of normal physiological requirements. They should be treated symptomatically, where the dosage being reduced or the drug slowly withdrawn.

Treatment of adverse effects

For gastrointestinal effects - Administration of antacids between meals may relieve indigestion or mild gastrointestinal irritation that may occur during parenteral therapy. However, the efficacy of antacids or other antilucer medications in preventing severe gastrointestinal problems, such as ulceration, hemorrhage, and/or bowel perforation, during corticosteroid therapy has not been established.

Mental depression or psychoses - If possible, decrease corticosteroid dosage or discontinue therapy. Phenothiazines may be administered if necessary. Lithium has been recommended. Some patients may require electroconvulsive therapy if severe depression persist. Tricyclic antidepressants should not be used since they do not relieve, and may exacerbate, corticosteroid-induced mental disturbances.

For withdrawal effects (non-HPA axis suppression) - Administration of aspirin or another NSAID may alleviate some of the symptoms of this condition.

Anaphylactic and hypersensitivity reactions may be treated with adrenaline, positive-pressure artificial respiration and aminophylline. The patient should be kept warm and quiet.

PACKING / PACK SIZES:

2 mL amber glass vials; 50 vials / carton

STORAGE CONDITIONS, USER INSTRUCTIONS AND PHARMACEUTICAL PRECAUTIONS:

Store below 30°C.

Protect from light and freezing. Sensitive to heat.

SHELF LIFE: 2 years

Manufacturer and Product Registration Holder

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SM PHARMACEUTICALS SDN BHD

DEXIN (DEXAMETHASONE PHOSPHATE) INJECTION 8MG/2ML

DESCRIPTION:

Clear, colourless solution in glass vials

COMPOSITION:

Each mL contains Dexamethasone Sodium Phosphate equivalent to Dexamethasone Phosphate 4 mg.

ACTIONS AND MODE OR MECHANISMS OF ACTION:

Dexamethasone is a highly potent and long-acting glucocorticoid with negligible sodium-retaining (mineralocorticoid) properties.

Corticosteroids - Diffuse across cell membranes and complex with specific cytoplasmic receptors. These complexes then enter the cell nucleus, bind to DNA, and stimulate transcription of messenger RNA (mRNA) and subsequent protein synthesis of various enzymes thought to be ultimately responsible for two categories of effects of systemic corticosteroids. However, these agents may suppress transcription of mRNA in some cells (e.g. lymphocytes).

Anti-inflammatory (steroidal) - Glucocorticoids decrease or prevent tissue responses to inflammatory processes, thereby reducing development of symptoms of inflammation without affecting the underlying cause. Glucocorticoids inhibit accumulation of inflammatory cells, including macrophages and leukocytes, at sites of inflammation. They also inhibit phagocytosis, lysosomal enzyme release, and synthesis and/or release of several chemical mediators of inflammation. Immunosuppressant actions may also contribute significantly to the anti-inflammatory effect.

Immunosuppressant - May involve prevention or suppression of cell-mediated (delayed hypersensitivity) immune reactions as well as more specific actions affecting the immune response. Glucocorticoids reduce the concentration of thymus-dependent lymphocytes (T-lymphocytes), monocytes, and eosinophils. They also decrease binding of immunoglobulin to cell surface receptors and inhibit the synthesis and/or release of interleukins, thereby decreasing T-lymphocyte blastogenesis and reducing expansion of the primary immune response. Glucocorticoids may also decrease passage of immune complexes through basement membranes and decrease concentrations of complement components and immunoglobulins.

Other actions / effects

Pharmacologic (supraphysiologic) doses of erogenous corticosteroids produce hypothalamic-pituitary-adrenal (HPA) axis suppression via a negative feedback mechanism, i.e., they inhibit pituitary ACTH secretion, thereby reducing ACTH-mediated production of corticosteroids and androgens in the adrenal cortex.

Glucocorticoids also stimulate protein catabolism and induce enzymes responsible for metabolism of amino acids; increase glucose availability by inducing hepatic enzymes involved in gluconeogenesis, stimulating protein catabolism and decreasing peripheral utilization of glucose; increase lipolysis and mobilize fatty acids from adipose tissues, leading to increased plasma fatty acid concentrations and decrease bone formation and increase bone resorption.

PHARMACOLOGY (SUMMARY OF PHARMACODYNAMICS AND PHARMACOKINETICS):

Absorption - Rapidly absorbed.

Distribution - Tissue distribution studies in animals show high uptake of dexamethasone by the liver, kidney and adrenal glands.

Bio-transformation - Primarily hepatic (slow and rather limited); also renal and tissue; mostly to inactive metabolites.

Duration of action - Depends upon the solubility of the dosage form as well as the biological (tissue) half-life (about 190 minutes).

Protein binding - Bound (up to 77%) to plasma proteins, mainly albumin.

Elimination - Over 60% of the administered dose is excreted in the urine within 24 hours, largely as un-conjugated steroids.

INDICATIONS:

Use in certain endocrine and non-endocrine disorders responsive to corticosteroid therapy.

Systemic Administration: Dexamethasone injection is recommended for systemic administration by IV or IM injection when oral therapy is not feasible or desirable in the following conditions.

Endocrine Disorders: Primary or secondary adrenocortical insufficiency. (Hydrocortisone or cortisone is the first choice, but synthetic analogues may be used with mineralocorticoids where applicable and in infancy, mineralocorticoid supplementation is particularly important.)

Non-Endocrine Disorders: Dexamethasone injection may be used in the treatment of non-endocrine corticosteroid responsive conditions including:

Allergy and Anaphylaxis: Angioneurotic oedema and anaphylaxis.

Gastrointestinal: Crohn's disease and ulcerative colitis. Infection (with appropriate chemotherapy). Miliary tuberculosis and endotoxic shock.

Neurological Disorders: Raised intracranial pressure secondary to tumours and infantile spasms.

Respiratory: Bronchial asthma and aspiration pneumonitis. Skin Disorders: Toxic epidermal necrolysis.

Shock: Adjunctive treatment where high pharmacological doses are needed. Treatment is an adjunct to, and not a substitute for specific and supportive measures the patient may require. Dexamethasone has been shown to be beneficial when used in the early treatment of shock, but it may not influence overall survival.

Local Administration: Dexamethasone injection is suitable for intra-articular or soft-tissue injection as adjunctive therapy for short-term administration in:

Soft tissue disorders, e.g. carpal tunnel syndrome and tenosynovitis.

Intra-articular disorders, e.g. rheumatoid arthritis and osteoarthritis with an inflammatory component.

Dexamethasone injection may be injected intra-lesionally in selected skin disorders, e.g. cystic acne vulgaris, localised lichen simplex and keloids.

CONTRAINDICATIONS:

Corticosteroids are usually contra-indicated in the presence of acute and systemic fungal infections. It is also contraindicated in patients with known hypersensitivity to components of this product.

SIDE EFFECTS / ADVERSE REACTIONS:

Fluid and Electrolyte Disturbances: Sodium retention, fluid retention, congestive heart failure in susceptible patients, potassium loss, hypokalaemic alkalosis, hypertension, increased calcium excretion.

Musculoskeletal: Muscle weakness, steroid myopathy, loss of muscle mass, osteoporosis (especially in post-menopausal females), vertebral compression fractures, aseptic necrosis of femoral and humeral heads, pathological fracture of long bones, tendon rupture and post-injection flare (following intra-articular use).

Gastrointestinal: Peptic ulcer with possible perforation and haemorrhage, perforation of the small and large bowel, particularly in patients with inflammatory bowel disease, pancreatitis, abdominal distension, ulcerative oesophagitis, dyspepsia, oesophageal candidiasis.

Dermatological: Impaired wound healing, thin fragile skin, petechiae and ecchymoses, erythema, striae, telangiectasia, acne, increased sweating, possible suppression of skin tests, burning or tingling especially in the perineal area (after IV injection), other cutaneous reactions, e.g. allergic dermatitis, urticaria, angio-neurotic oedema and hypo-or hyper-pigmentation.

Neurological: Convulsions, increased intra-cranial pressure with papillo-oedema (pseudotumour cerebri) usually after treatment, vertigo, headache and rare instances of blindness associated with intra-lesional therapy around the face and head.

Endocrine: Menstrual irregularities, amenorrhoea, development of Cushingoid state, suppression of growth in children, secondary adreno-cortical and pituitary unresponsiveness, particularly in times of stress, as in trauma, surgery or illness, decreased carbohydrate tolerance, manifestations of latent diabetes mellitus, increased requirements for insulin or oral hypoglycaemic agents in diabetes. Ophthalmic: Posterior sub-capsular cataracts, increased intra-ocular, papillo-oedema, corneal or scleral thinning, exacerbation of ophthalmic viral diseases, glaucoma, exophthalmos.

Metabolic: Negative nitrogen balance due to protein catabolism.

Others: Anaphylactoid or hypersensitivity reactions, leucocytosis, thrombo-embolism, weight gain, increased appetite, nausea, malaise and sterile abscess.

PRECAUTIONS / WARNINGS:

Unless considered life saving, systemic administration of corticosteroids is contraindicated in patients with peptic ulcer, osteoporosis, psychoses, or severe psychoneuroses and they should be used only with great caution in the presence of congestive heart failure or hypertension, in patients with diabetes mellitus, epilepsy, glaucoma, infectious diseases, ocular herpes simplex, chronic renal failure, and uraemia, and in elderly persons. Patients with active or doubtfully quiescent tuberculosis should not be given corticosteroids except, very rarely, as adjuncts to treatment with anti-tubercular drugs. Patients with quiescent tuberculosis should be observed closely and should receive chemoprophylaxis if corticosteroid therapy is prolonged.

Withdrawal symptoms

Stopping corticosteroids after prolonged therapy may cause withdrawal symptoms including fever, myalgia, arthralgia and malaise. This may occur in patients even without evidence of adrenal insufficiency.

Psychiatric disturbances

Psychic derangements may appear when corticosteroids are used. These may range from euphoria, insomnia, mood swings, personality changes and severe depression, to frank psychotic manifestation. Existing emotional instability or psychotic tendencies may also be aggravated by corticosteroids.

Intra-articular injection

This medication should not be used when the following medical problems exist: prior to arthro-plasty of joints (increased risk of infection); blood clotting disorders (risk of intra - and extra-articular haemorrhage); intra-articular fracture (healing may be retarded); current or history of peri-articular infection (may be exacerbated or reactivated); non-arthritis juxta-articular osteoporosis (may be exacerbated) and unstable joint.

Frequent intra-articular injections have been reported to cause development of Charcot-like arthro-pathies.

Warnings:

Following intra-articular injection, the injected joints should not be overused even if pain is relieved, because of the increased risk of joint damage or deterioration. It is recommended that weight-bearing joints be rested for 24 to 48 hours post-injection.

Myasthenia gravis

Muscle weakness may be increased initially, possibly leading to respiratory distress; patients should be hospitalised, and respiratory support should be immediately available, when glucocorticoid therapy is initiated.

High Risk Groups

Pregnancy

Corticosteroids cross the placenta. Although adequate studies have not been done in humans, there is some evidence that pharmacologic doses of corticosteroids may increase the risk of placental insufficiency, decreased birth weight, or stillbirth. However, teratogenic effects in humans have not been confirmed. Infants born to mothers who have received substantial doses of corticosteroids during pregnancy should be carefully observed for signs of hypoadrenalism and replacement therapy administered if required.

Breast feeding

Problems in humans have not been documented. Breast-feeding is not recommended because it is excreted in breast milk and may cause unwanted effects, such as growth suppression and inhibition of endogenous steroid production, in the infant.

Paediatrics

Chronic use of corticosteroids may suppress growth and development of the paediatric or adolescent patient and should be undertaken with caution. Also, paediatric patients may be at increased risk of developing osteoporosis, avascular necrosis of the femoral heads, glaucoma, or cataracts during prolonged therapy. Children and adolescents receiving prolonged therapy should be closely monitored. Large doses of corticosteroids may be associated with intracranial hypertension in children.

Concurrent disease

Caution should be exercised in administering large doses of dexamethasone to patients with known cardiac failure, hypertension, peptic ulceration, glaucoma, epilepsy, diabetes mellitus, chronic psychosis, and a past history of tuberculosis. The effects are enhanced in hepatic disease and hypothyroidism.

WARNING: SODIUM METABISULPHITE.

This product contains Sodium Metabisulphite that may cause serious allergic type reactions in certain susceptible patients. Do not use if known to be hypersensitive to bisulphites.

Pheochromocytoma crisis, which may be fatal, has been reported after administration of systemic corticosteroids. Corticosteroids should only be administered to patients with suspected or identified pheochromocytoma after an appropriate risk/benefit evaluation.

DRUG INTERACTIONS:

Acetaminophen

Induction of hepatic enzymes by corticosteroids may increase the formation of a hepatotoxic acetaminophen metabolite, thereby increasing the risk of hepatotoxicity, when they are used concurrently with chronic or high-dose acetaminophen therapy.

Alcohol or Anti-inflammatory drugs, nonsteroidal (NSAIDs)

Risk of gastrointestinal ulceration or hemorrhage may be increased when these substances are used concurrently with glucocorticoids; however, concurrent use of NSAIDs in the treatment of arthritis may provide additive therapeutic benefit and permit glucocorticoid dosage reduction.

Aminoglutethimide

Aminoglutethimide suppresses adrenal function so that glucocorticoid supplementation may be required

Amphotericin B, parenteral or Carbonic anhydrase inhibitors

Concurrent use with corticosteroids may result in severe hypokalaemia and should be undertaken with caution; serum potassium concentrations and cardiac function should be monitored during concurrent use.

The possibility should be considered that concurrent chronic use of both carbonic anhydrase inhibitors and corticosteroids may increase the risk of hypocalcaemia and osteoporosis because carbonic anhydrase inhibitors also increase calcium excretion.

Anabolic steroids or Androgens

Concurrent use with glucocorticoids may increase the risk of oedema. Also, concurrent use may promote the development of severe acne.

Antacids

Concurrent chronic use with dexamethasone may decrease absorption of this glucocorticoid; efficacy may be decreased sufficiently to require dosage adjustment in patients receiving small doses, but probably not in those receiving large doses, of the corticosteroid.

Anticholinergics, especially atropine and related compounds

Concurrent long-term use with glucocorticoids may increase intraocular pressure.

Anticoagulants, coumarin, or indandione - derivative, or Heparin or Streptokinase or Urokinase

Effects of coumarin or indandione derivatives are usually decreased but may be increased in some patients when used concurrently with glucocorticoids. Dosage adjustments based on prothrombin time determinations may be necessary during and after glucocorticoid therapy.

The potential occurrence of gastrointestinal ulceration or hemorrhage during glucocorticoid therapy and the effects of glucocorticoid on vascular integrity, may cause increased risk.

Antidepressants, tricyclic

These medications do not relieve, and may exacerbate corticosteroid-induced mental disturbances. They should not be used for treatment of these adverse effects.

Antidiabetic agents, sulfonylurea or Insulin

Glucocorticoids may increase blood glucose concentration; dosage adjustment of one or both agents may be necessary during concurrent use.

Anticholinergics, especially atropine and related compounds

Concurrent long-term use with glucocorticoids may increase intraocular pressure.

Contraceptives, oral, estrogen-containing, or Oestrogens

Oestrogens may alter the metabolism and protein binding of glucocorticoids, leading to decreased clearance, increased elimination half - life, and increased therapeutic and toxic effects of the glucocorticoid.

Digitalis glycosides

Concurrent use with glucocorticoids may increase the possibility of arrhythmias or digitalis toxicity associated with hypokalaemia.

Diuretics

Sodium and fluid - retaining actions of corticosteroids may decrease natriuretic and diuretic effects of these medications and vice versa.

Concurrent use of potassium - depleting diuretics with corticosteroids may result in severe hypokalaemia.

Effects of potassium - sparing diuretics and / or corticosteroids on serum potassium concentration may be decreased during concurrent use.

Ephedrine, phenobarbitone, phenytoin, rifampicin

Concurrent use may decrease the corticosteroid effect because of increased corticosteroid metabolism resulting from induction of hepatic microsomal enzymes.

Folic Acid

Requirements may be increased in patients receiving long-term corticosteroid therapy.

Immunosuppressant agents, other

Concurrent use with immunosuppressant doses of glucocorticoid may increase the risk of infection and development of lymphomas or other lymphoproliferative disorders.

Isoniazid, Mexiletine

Concurrent use with glucocorticoids may increase hepatic metabolism of isoniazid and mexiletine and / or excretion of isoniazid, leading to decreased plasma concentration of the drugs.

Iophendylate or Metrizamide

Concurrent intrathecal administration of metrizamide or iophendylate with intrathecal administration of glucocorticoids may increase the risk of arachnoiditis.

Mitotane

Glucocorticoid supplementation is usually required during mitotane administrations, but higher doses than those generally used for replacement therapy may be required because mitotane alters glucocorticoid metabolism.